

The role of virgin olive oil components in the modulation of endothelial function.



The endothelium is involved in many of the processes related to the development of atherosclerosis, which is considered an inflammatory disease. Actually, traditional risk factors for atherosclerosis predispose to endothelial dysfunction, which is manifested as an increase in the expression of specific cytokines and adhesion molecules. There are firm evidence supporting the beneficial effects of olive oil, the most genuine component of the Mediterranean diet. Although the effects of olive oil and other oleic acid-rich dietary oils on atherosclerosis and plasma lipids are well known, the roles of minor components have been less investigated. Minor components constitute only 1-2% of virgin olive oil (VOO) and are composed of hydrocarbons, polyphenols, tocopherols, sterols, triterpenoids and other components usually found in traces. Despite their low concentration, non-fatty acid constituents may be of importance because studies comparing monounsaturated dietary oils have reported different effects on cardiovascular disease. Most of these compounds have demonstrated antioxidant, anti-inflammatory and hypolipidemic properties. In this review, we summarize current knowledge on the effects of these compounds contained in VOO on vascular dysfunction and the mechanisms by which they modulate endothelial activity. Such mechanisms involve the release of nitric oxide, eicosanoids (prostaglandins and leukotrienes) and adhesion molecules, in most cases by activation of nuclear factor kappaB by reactive oxygen species.